

Discovery Executive Summary

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EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 2 discovery analyses run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating	Hotspot Score
1	Testing & Quality Assurance Analysis	HIGH RISK	—
2	Technical Debt	MODERATE	—

1. Testing & Quality Assurance Analysis

OVERALL CODEBASE RATING — TESTING & QUALITY ASSURANCE

High Risk

Driven by zero-tested critical logic (H1), ~2% coverage (H2), and the total absence of integration, contract, frontend and E2E tests (H3, H4, H7).

EXECUTIVE SUMMARY

The repository ships with a test suite that exists in name only: **2 PHPUnit files** guard a codebase of **36 source files** across three layers (Laravel backend, Node/Express `dev-api`, React frontend). Both existing tests are vacuous — one asserts a hard-coded literal array equals itself, the other loops over `random_int()` and asserts a probabilistic threshold — so the **effective coverage of real business logic is ~2% (estimated)**; there is no coverage tooling (`coverage: none` in CI) to measure it. Every business-critical module is untested: the IVR-traversal / reachability engine (`RealTimeTestService`, `dev-api/src/realtime.js`), the Mongo persistence layer (`MongoService`), the unsafe-`extract()` `LegacyDataMapper`, and all six API controllers exposing ~18 endpoints. There are **no integration tests** exercising the DB/Mongo/HTTP/SSE boundaries, **no contract tests** for any public API, and **no frontend or end-to-end tests at all** despite a React app with routing, forms, and live SSE feeds. CI runs `phpunit` and a frontend `build` on every PR, but with the suite this thin the gate provides false confidence rather than protection. Overall testing health is **High Risk** and must be raised before any refactor or extraction work begins.

5.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	Untested Critical Logic	Critical modules with zero tests	0	1–3	>3	6+ (RealTimeTestService, MongoService, LegacyDataMapper, ConnectController, DiscoveryController, dev-api/realtime.js)	HIGH RISK
H2	Low Test Coverage	Overall coverage %	>80%	50–80%	<50%	~2% (est., no coverage tool)	HIGH RISK
H3	Missing Integration Tests	Boundaries covered %	>70%	30–70%	<30%	0% (no Feature/HTTP/DB/SSE tests)	HIGH RISK
H4	Missing Contract Tests	APIs with contract tests %	>80%	40–80%	<40%	0% (~18 endpoints, 0 covered)	HIGH RISK
H5	Flaky / Skipped Tests	Skipped/flaky test count	0	1–5	>5	1 (non-deterministic <code>random_int</code> test)	MODERATE
H6	No CI Test Gate	Tests enforced in CI	Required gate	Runs, not required	No CI test run	Backend phpunit runs on PR (not proven required); frontend runs no tests	MODERATE
H7	No End-to-End Tests (additional)	Critical user flows with E2E coverage %	>60%	20–60%	<20%	0% (no Cypress/Playwright)	HIGH RISK
H8	Assertion-Free / Vacuous Tests (additional)	Tests with meaningful assertions %	>90%	50–90%	<50%	0% (0 of 2 assert real logic)	HIGH RISK

5.4 Actions Required

Hotspot	Action	Rating	Priority
H1 Untested Critical Logic	Add PHPUnit unit tests for <code>RealTimeTestService</code> , the reachability calc, <code>LegacyDataMapper</code> , <code>MongoService</code> ; mirror in <code>dev-api</code>	HIGH RISK	CRITICAL
H2 Low Test Coverage	Enable coverage tooling per layer; drive critical modules to 75–80% before refactors	HIGH RISK	CRITICAL

H3 Missing Integration Tests	Add Laravel feature tests (<code>RefreshDatabase</code>) + Mongo round-trip (<code>mongodb-memory-server</code>) + SSE smoke tests	HIGH RISK	HIGH
H4 Missing Contract Tests	Add <code>assertJsonStructure</code> tests for all ~18 API routes; validate frontend client against <code>types/index.ts</code>	HIGH RISK	HIGH
H7 No End-to-End Tests	Add Playwright happy-path specs per module (discovery + connect flows), run against <code>dev-api</code> in CI	HIGH RISK	HIGH
H8 Vacuous Tests	Replace both tautological tests with real assertions; add mutation testing (Infection)	HIGH RISK	HIGH
H5 Flaky Test	Delete the <code>random_int</code> reachability test; ban randomness in <code>tests/</code> ; replace with deterministic case	MODERATE	MEDIUM
H6 No CI Test Gate	Add frontend test step, enable coverage, make test jobs required branch-protection checks	MODERATE	MEDIUM

5.5 Expected Outcomes

- **Critical telecom logic is protected before refactors** — IVR traversal, reachability scoring, and the unsafe `LegacyDataMapper` gain real unit tests, so extraction/refactor work has a behavioural safety net.
- **Measurable coverage replaces guesswork** — per-layer coverage instruments turn "~2% estimated" into an enforced, ratcheting number on the path to 75–80%.
- **Boundaries are verified, not assumed** — integration tests exercise the controller → MariaDB → Mongo → SSE wiring, catching failures that unit tests miss.
- **API contracts can't drift silently** — contract tests on all ~18 endpoints (and the frontend client) fail the build the moment a response shape changes, protecting the customer dashboard.
- **The frontend gains a safety net** — Vitest unit tests plus Playwright E2E cover the React flows and live SSE feed that today have zero automated verification.
- **CI becomes a real gate** — a required, coverage-aware check across backend, frontend, and E2E means regressions are blocked at merge instead of discovered in production.

The complete report (including §5.2 Hotspot-by-Hotspot Evidence with `affected-files` directives and §5.3 Mermaid diagrams) is saved at `docs/discovery/05-testing-and-quality-assurance.md`, ready for the UI to render to PDF.

2. Technical Debt

OVERALL CODEBASE RATING — TECHNICAL DEBT & AGENTIC READINESS

Moderate

Driven by a missing `composer.lock` (non-reproducible backend builds), no enforced lint/style gate, no DB migration framework, and shallow/non-deterministic test coverage — no single dimension is High Risk, but every dimension carries real gaps.

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Klearcom is a well-structured polyglot monolith with genuinely strong agentic scaffolding: five `AGENTS.md` guides, a symmetric Discovery/Connect module layout, a working GitHub Actions CI pipeline, and Docker Compose for the full stack. That foundation makes it more agent-ready than most codebases of its age. The debt is concentrated in reproducibility and quality-gate depth rather than architecture. The three most severe gaps are: (1) `backend/composer.lock` is not committed, so every CI run and every developer install resolves PHP dependencies afresh — backend builds are not reproducible; (2) **no code-style enforcement exists** — no ESLint, Prettier, Laravel Pint, and although `phpstan/phpstan` is declared it has no config file and is never invoked in CI, so agent-authored code has no automated style/lint gate; (3) **the database has no migration framework** — the entire schema lives in a single `docker/mariadb/init.sql` that only executes on a fresh MariaDB volume, leaving no versioned, reversible path for schema evolution. There are also self-documented debts (a deliberately reintroduced `extract()` pattern, a non-deterministic test using `random_int()`, missing dev-api input validation and rate limiting) captured in `docs/CODEBASE_AUDIT_ISSUES.md`. Overall the repo is **Moderate** readiness: a real CI gate exists and can be trusted for the frontend, but the missing lock file, absent lint layer, and flaky/shallow tests mean agent-authored output cannot yet be verified with full confidence.

Readiness Benchmark Ratings

#	Dimension	GOOD	MODERATE	HIGH RISK	Measured	R
D1	Code Repository Health	all checks pass	1–2 gaps	3+ gaps / no CI	<code>.gitignore</code> + CI + JS lock files present; <code>composer.lock</code> missing, no CODEOWNERS/PR template, CI runs no lint step	
D2	Third-Party Tool Usage	mostly wired & current	some unused/unwired	many unused/unmaintained	18/19 packages wired; <code>phpstan</code> declared but no config and never run	
D3	AI Tool / Agentic Readiness	ready	partial	not ready	5 <code>AGENTS.md</code> + <code>.kiro/</code> config + symmetric enumerable modules; but CI verification gate is shallow (no lint, one flaky test, dev-api untested)	
D4	Database Usage	sound	some gaps	no constraints / shared flat schema	FKs with <code>ON DELETE CASCADE</code> + Mongo indexes present; no migration framework , few secondary indexes, schema+seed combined & non-idempotent	
D5	Development Environment	reproducible	partial	manual / fragile	Docker Compose + Dockerfiles + backend/dev-api <code>.env.example</code> ; no lint/formatter enforced , no <code>frontend/.env.example</code>	
D6	Automated Quality Gates & Test Determinism (additional)	lint+type+deterministic tests gate every PR	partial gate	no gate / flaky	CI runs <code>phpunit</code> + <code>tsc --noEmit</code> + <code>vite build</code> ; but no lint, no coverage, <code>ReachabilityCalculationTest</code>	

uses `random_int()` (flaky), dev-api has zero tests

Additional readiness gaps: one additional dimension (D6 — Automated Quality Gates & Test Determinism) was observed beyond the five standard dimensions, driven by the flaky `random_int()` test and the absence of any lint step in CI. No further readiness gaps beyond these were observed.

8.8 Actions Required

Gap	Action	Rating	Priority
<code>backend/composer.lock</code> not committed — non-reproducible backend builds	Run <code>composer update</code> and commit <code>composer.lock</code> ; keep it in sync going forward	MODERATE	CRITICAL
No code-style / static-analysis enforcement	Add ESLint + Prettier (frontend), Laravel Pint + a <code>phpstan.neon</code> (backend); wire all into CI and a pre-commit hook	MODERATE	HIGH
CI gate too shallow to accept agent output	Add a lint/static-analysis job, a <code>dev-api</code> test job, and make <code>ReachabilityCalculationTest</code> deterministic (remove <code>random_int()</code>)	MODERATE	HIGH
No database migration framework — schema only applies on fresh volume	Adopt Laravel migrations (or a versioned SQL tool); back-fill current schema as migration 0001; add rollback path	MODERATE	HIGH
<code>phpstan</code> declared but never configured or run	Add <code>phpstan.neon</code> and a <code>phpstan analyse</code> CI step, or remove the dependency	MODERATE	MEDIUM
No branch-protection signals (CODEOWNERS, PR template)	Add <code>CODEOWNERS</code> per top-level module and <code>.github/PULL_REQUEST_TEMPLATE.md</code> ; require CI checks on <code>main</code>	MODERATE	MEDIUM
Missing secondary DB indexes on FK / status / lookup columns	Add indexes on <code>discovery_nodes(discovery_job_id, parent_id)</code> , <code>connect_check_results(connect_monitor_id)</code> , and <code>*.status</code>	MODERATE	MEDIUM
Schema DDL and seed data combined; container seeds non-idempotent	Split DDL from seed data; guard seed inserts behind existence checks	MODERATE	LOW
No <code>frontend/.env.example</code> ; <code>frontend/Dockerfile</code> uses <code>npm install</code> not <code>npm ci</code>	Add <code>frontend/.env.example</code> with <code>VITE_API_URL</code> ; switch Dockerfile to <code>npm ci</code>	MODERATE	LOW

The Markdown deliverable is written; the orchestration UI will render `docs/discovery/08-technical-debt.pdf` from it.